

The University of Nottingham

SCHOOL OF Computer Science

A LEVEL THREE MODULE, SPRING SEMESTER 2020-2021

Computer Vision

Time allowed TWO HOURS

Answer THREE out of FOUR Questions***Each question carries 20 marks*****ADDITIONAL MATERIAL:** Nil.**INSTRUCTION FOR STUDENTS:**

- The university's honour code applies to this individual short time assessment. The submitted answers **must be your own work** where **answers are attempted without any help from others**. University academic misconduct rules apply here in the case of plagiarism, collusion or false authorship.
- Write the name and student ID at the top of your answer sheets.
- Students are not allowed to ask questions during the short time assessment in accordance with the no shout-out policy.
- You may answer the questions in any order but write the question numbers correctly and clearly.
- At the end of the exam, **upload your** solutions and name the file as **StudentID**. Upload your solution file onto **Moodle**.
- Please ensure that you upload the correct files to the folder labelled as "Final examination submission folder" in Moodle. Request for resubmission will not be accepted after the submission.
- Students should stop writing the answers at the end of exam time. Answer scripts must be submitted to Moodle within **thirty minutes** after the end of exam. In case of any technical issues, the students need to contact the module convener in this period.

IMPORTANT: Late submission will not be accepted.

1. The following questions are pertaining to Features, Image Segmentation, and Object Detection.

a. Image segmentation is a process of partitioning image pixels into different groups for better understanding and/or easier analysis of the image. k-means is a typical clustering algorithm that has been widely applied to image segmentation.

i. The objective function of k-means algorithm is typically formulated as follows

$$L = \sum_{i=1}^N \sum_{k=1}^K r_{ik} \|x_i - \mu_k\|^2 \quad (1)$$

where μ_k is the centre of the k -th cluster, x_i is the feature vector of i -th data point, and r_{ik} the membership of data point i belonging to cluster k . With respect to the input image in Figure 1, assuming the image is to be segmented into three categories including chair, table, and background, specify what x_i , μ_k , and N refer to respectively, and explain whether colour is a good choice of features for the segmentation task.

[3 marks]

ii. K-means algorithm is derived by minimising the objective function (1) with respect to the unknown parameters μ_k and r_{ik} . Based on (1), explain how k-means would perform for the instance segmentation task as shown in Figure 1, provided that a feature vector is represented by pixel coordinates.

[4 marks]

iii. As the objective function (1) does not explicitly consider the neighbourhood structure between image pixels, a researcher tries to modify the objective function as follows:

$$L = \sum_{i=1}^N \sum_{k=1}^K [r_{ik} \|x_i - \mu_k\|^2 + \sum_{j \in \Omega(i)} f(x_i, x_j)] \quad (2)$$

where $\Omega(i)$ is the neighbourhood of the pixel indexed by i , $f(x_i, x_j)$ is a function only depending on the value of feature vectors of pixel i and j . What effect would the k-means algorithm derived by minimising the modified objective function (2) have compared to that derived by minimising the objective function (1)? Explain.

[3 marks]

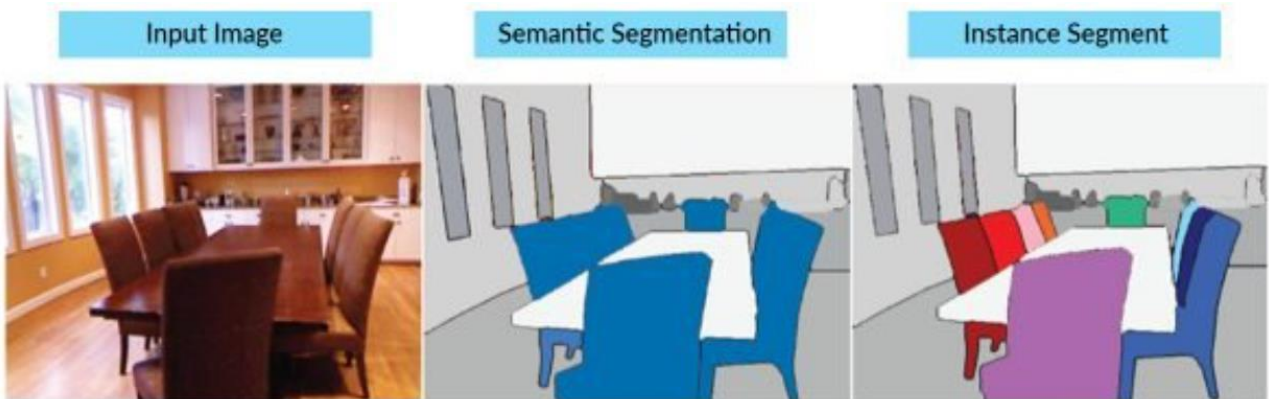


Figure 1

- b. To perform the instance segmentation task for chairs as demonstrated in Figure 1, a researcher designed a two-stage method, i.e., to detect a chair by fitting a pictorial template to the image and then to refine the labelling of pixels within the template. Assume the researcher is using a graph $G = \langle V, E \rangle$ to represent the pictorial template shown in Figure 2, where V and E represent the set of parts and the relations between a pair of parts, respectively.

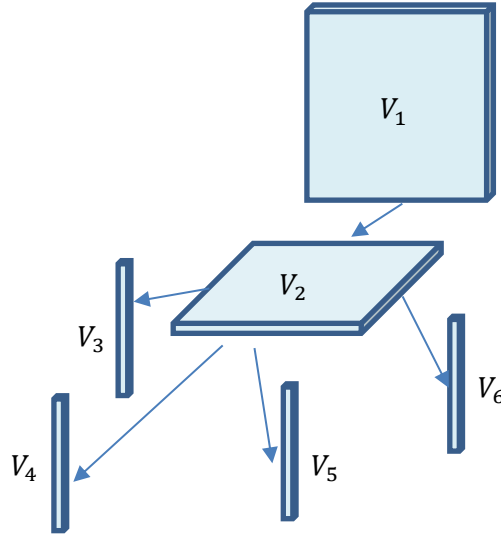


Figure 2

- i. Describe a set of measurable features of a part and explain why the researcher should consider them in the graph. [2 marks]
- ii. To fit the pictorial model in the input image, the cost function of $L = \sum_i C(V_i) + \sum_{(i,j) \in E} C(V_i, V_j)$. Explain what $C(V_i)$ and $C(V_i, V_j)$ measure in the pictorial model. [2 marks]
- iii. What is the search space to fit the pictorial model to the input image in Figure 1? [3 marks]
- iv. What challenges would the pictorial structure based method have in detecting individual chairs in Figure 1? Explain. [3 marks]

2. The following questions are pertaining to Stereo Vision.

- a. Both Epipolar Geometry and Disparity Gradient have been exploited in several Stereo Vision algorithms to find the correspondence between pixels in left and right images.
- Explain Disparity Gradient, Disparity Gradient constraint, and how Disparity Gradient constraint is transformed into a constraint on viewed surfaces, respectively. [3 marks]
 - Explain with sketch how Epipolar Geometry constrains the search of the corresponding pixel in the right image for a pixel in the left image. [3 marks]
 - Describe conceptually how Epipolar Geometry constraint and Disparity Gradient constraint are combined together in Zhang and Shan's progressive scheme for searching stereo correspondences. [4 marks]
- b. Scanline method is one of the global optimisation methods that solves stereo correspondence problem by minimising the combined cost of individual pixel assignment as well as pairwise pixel assignment.
- Briefly describe the possible assignments to a pixel and explain their meanings respectively. [3 marks]
 - To apply scanline method, stereo correspondence must satisfy ordering constraint. Explain what ordering constraint means and illustrate an example where ordering constraint is violated. [3 marks]
 - Given the following scenario of left scanline and right scanline, each square representing a pixel. Assume two pixels (blue colour) in the left scanline and one pixel (red colour) in the right scanline are occluded. Sketch a possible stereo correspondence solution that satisfies the ordering constraint. Justify your answer. [4 marks]

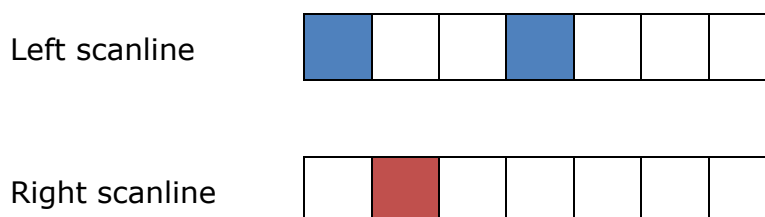


Figure 3

3. The following questions are pertaining to Optical Flow and Visual Tracking.

- a. Both Horn-Schunck and Lucas Kanade algorithms are designed for optical flow problems.
- Explain why Horn-Schunck is a global algorithm whilst Lucas-Kanade is a local algorithm. [2 marks]
 - Briefly describe how Lucas-Kanade algorithm employs image pyramid to address the issue of large motion in optical flow. [5 marks]
 - Figure 4 shows an opaque square object (gray colour) is moving to the top-right. Explain if the motion at three apertures (a), (b), and (c) can be fully recovered. Justify your answer. [3 marks]

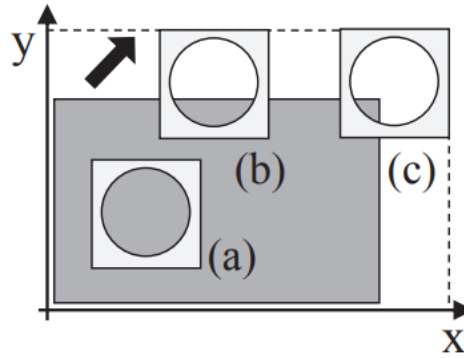


Figure 4

[3 marks]

- b. CONDENSATION is an effective algorithm for real tracking. It implements particle filters that is well grounded in Bayes Theorem which formulates that

$$P(x_{t+1} | \mathbf{Z}_{t+1}) = k_{t+1} \cdot P(z_{t+1} | x_{t+1})P(x_{t+1} | \mathbf{Z}_t) \quad (1)$$

$$P(x_{t+1} | \mathbf{Z}_t) = \int_{x_t} P(x_{t+1} | x_t)P(x_t | \mathbf{Z}_t) \quad (2)$$

$P(\cdot)$ represents a probability function; x_t is the state of target at time t and its total history (from time 0 to t) is $\mathbf{X}_t = \{x_1, x_2, \dots, x_t\}$; z_t is the set of image features detected at time t and its total history is $\mathbf{Z}_t = \{z_1, z_2, \dots, z_t\}$; and $P(x_{t+1} | \mathbf{X}_t) = P(x_{t+1} | x_t)$.

- As there is no explicit representation of $P(x_t | \mathbf{Z}_t)$, it is approximated by a set of discrete particles in CONDENSATION. Explain what each particle represents and whether the particles are tracking a single target or multiple targets. Justify your answer. [3 marks]

- ii. Explain the meaning of $P(z_{t+1}|x_{t+1})$ in equation (1) and how it is used in CONDENSATION algorithm. [2 marks]

- iii. Explain how equation (2) is approximated in CONDENSATION algorithm and why it is a plausible approximation. Justify your answer. [5 marks]

4. The following questions are pertaining to Convolutional Neural Networks.

- a. Convolutional Neural Networks (CNNs) have been widely applied to various Computer Vision problems. A CNN for image classification typically consists of common patterns of operations as shown in Figure 5, which demonstrates a CNN with two convolutional layers and a classifier. "Filter Bank" refers to a set of filtering operations that are to be applied to the (normalised) output of a previous layer. "Feature Pooling" refers to a subsampling process.

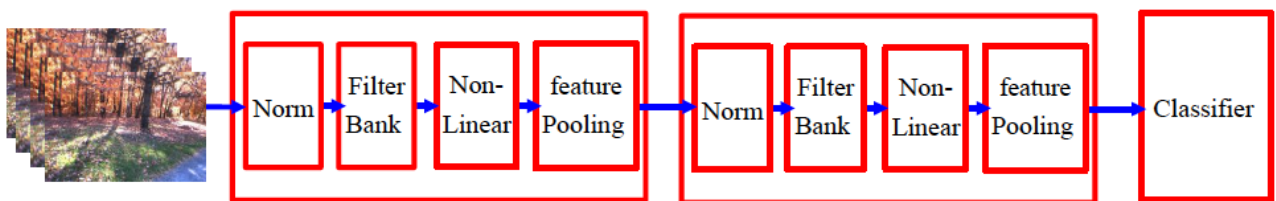


Figure 5

- i. Explain why a filtering process is considered a 3D operation whilst a feature pooling process is 2D. [2 marks]
- ii. Assume the input to the CNN is a colour image consisting of red, green, and blue channels and is of size 256×256 . Assume there are 64 filters in the filter bank in both convolutional layers, with filter size 5×5 in the first convolutional layer (taking in the input) and 3×3 in the latter convolutional layer (outputting to Classifier). Ignoring the bias, calculate the number of parameters in convolutional layers and explain why CNNs typically require large number of labelled samples for training. [4 marks]
- iii. Assuming the Classifier in the CNN in Figure 5 consists of fully connected layers, explain conceptually how the CNN can be trained. [4 marks]

b. There have been many advances with CNNs and some of them have significant impact in the development of Computer Vision field than others, including LeNet, AlexNet, VGG, GoogLeNet, ResNet, to name a few.

i. In conventional CNNs the filter size across different filters in the same convolutional layer must be the same. Explain how Inception unit proposed in GoogLeNet has overcome the constraint and the advantage of inception unit over a standard convolutional layer.

[3 marks]

ii. ResNet implements skip connections as shown in Figure 6 to address the problem of vanishing gradient in training deep CNNs. Explain, with respect to Figure 6, how the vanishing gradient problem is addressed and why it works.

[3 marks]

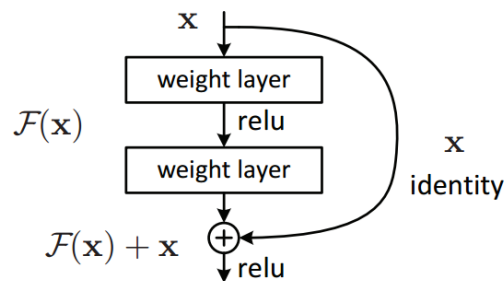


Figure 6

iii. Transfer learning is a popular technique when dealing with Computer Vision problems with relatively smaller sample size compared to that, for example, of ImageNet. Demonstrate with sketches as to how a VGG trained on ImageNet for classification tasks can be used through transfer learning for an image segmentation task. Justify your answer.

[4 marks]